

R&D Factors — Software & Services

Industry-specific, non-conventional equity factors built from the *R&D behavior* of software companies

Experiment 2 extension. Universe: GICS Software & Services. Sample: 1998-01 → 2026-01 (337 months, ~270–340 stocks/month). Tested under the Experiment 2 pipeline (even-quintile sorts + monthly Fama–MacBeth regressions, industry-neutral long/short alpha, turnover-cost model). All fundamentals are point-in-time (attached on `observation_date`, no look-ahead). Outputs: `RD/`.

0. The idea in one paragraph

The project already prices three R&D-*adjusted* characteristics — `intangible_value`, `intangible_profitability` and `rd_productivity` — but all three are **levels / valuation** signals built on the slow-moving accumulated R&D capital stock `K_int`. None of them measures the **dynamics and discipline of R&D spending itself**. This note proposes four factors that target four distinct *behavioral* dimensions of R&D the existing book is blind to — **flow, profit-output, consistency, and composition** — plus the conventional R&D-intensity *level* retained only as a baseline for contrast. The thesis we set out to test: *within a single R&D-intensive industry, how a firm manages R&D carries more cross-sectional information than how much R&D it has*.

#	Factor	R&D-behavior dimension	Definition	Hypothesised direction
1	<code>rd_growth</code>	Flow (is R&D ramping?)	$\Delta(\text{R\&D, YoY}) / \text{avg assets}$	long high
2	<code>rd_conversion</code>	Profit output (does R&D convert to margin?)	$\Delta(\text{gross income, YoY}) / \text{sales}_{(t-12)}$	long high
3	<code>rd_stability</code>	Consistency (is the R&D programme disciplined?)	– trailing-36m coeff. of variation of (R&D/sales)	long high
4	<code>innovation_mix</code>	Composition (build vs sell?)	$\text{R\&D} / (\text{R\&D} + \text{SG\&A})$	long high
5	<code>rd_intensity</code> (baseline)	Level (how much R&D?)	$\text{R\&D} / \text{sales}$	long high

Why these are non-conventional. Conventional R&D factors stop at the *level* of R&D (R&D/sales or R&D/market-cap). Three of our four are time-derivatives or moments of R&D behaviour rather than levels — and one of them (`rd_stability`) is a **second moment**, a type of signal that appears *nowhere* in the existing 15-factor book. They are all "straightforward" in the sense the brief asks: each is a single ratio, a year-over-year change, or one trailing-window moment computed from fields already in the data — no multi-step estimation.

1. Why R&D behavior should carry a premium — and why it survives

Everything below rests on **one permanent accounting fact** and **one permanent behavioral fact**:

- **Accounting (permanent).** Under ASC 730 / IAS 38, R&D is **expensed**, not capitalised. A dollar of R&D hits the income statement immediately and *never* appears as a balance-sheet asset with a visible depreciation schedule. So R&D-active firms' GAAP earnings, assets, and book value systematically *understate* economic reality, and conventional screens (P/E, P/B, ROA, asset growth) misread them. This treatment has been stable for decades and is not on any standard-setter's agenda.
- **Behavior (permanent).** Investors anchor on reported earnings ("functional fixation"); the sell-side in software fixates on revenue/bookings. Both under-weight investment that arrives as an *expense* and over-weight revenue that is *bought* rather than *earned*.

Together these produce mis-pricings that (i) **renew with every technology wave** (each wave creates a fresh cohort of R&D-ramping firms), (ii) **resist arbitrage** because R&D payoffs are multi-year, noisy, and require holding names with optically poor near-term profitability through the wait — career-risky and drawdown-prone, so fast money won't carry them, and (iii) **are not in vendor factor libraries**, which are overwhelmingly point-in-time cross-sectional ratios, not panel-constructed changes or moments. That is the structural reason these signals can persist without being competed away.

The four behavioral factors each isolate a *different* face of this mis-pricing. The sections below give, for each: the **predicting logic**, **why it stays robust**, **why it is not a re-skin of an existing factor**, **why it persists / has not been arbitrated**, and an **ex-ante prediction of sign and magnitude**.

2. The factors (ex-ante hypotheses)

Throughout, the long/short book is signed by the **hypothesised** direction (`higher_is_bullish = True` for all five), *not* by the in-sample t-stat. So a *negative* realised alpha t-stat means the factor worked **against** the hypothesis — the comparison in §4 reads directly off the sign.

2.1 `rd_growth` — the flow: is R&D ramping? (*long high*)

Definition. $(rd_ltm - rd_ltm_{(t-12)}) / \frac{1}{2}(assets + assets_{(t-12)})$ — the year-over-year change in R&D spending, scaled by average total assets. This is the **intangible-investment analogue of `asset_growth`**, built to be directly contrastable with it.

Predicting logic. A firm raising R&D/assets is voluntarily depressing current GAAP earnings to build off-balance-sheet intangible capital whose payoff is deferred. Earnings-anchored investors under-react to investment that shows up as an *expense* rather than a capitalised asset. So accelerating "disguised investment" should mark firms whose near-term earnings most understate economic earnings — and earn positive abnormal returns as the payoff arrives (Eberhart, Maxwell & Siddique 2004 document multi-year positive abnormal returns after large R&D increases).

Why robust. Rests on the permanent expensing rule plus permanent earnings-fixation; it is a ratio (currency-, size-, inflation-neutral) defined for every R&D reporter, and self-renewing across waves.

Why not correlated with existing factors. The crucial point is the **sign flip vs `asset_growth`** : tangible asset/capex growth predicts *low* returns (the investment anomaly), but R&D is expensed and **never enters `assets`** , so `rd_growth` captures exactly the investment that `asset_growth` is structurally blind to — with the *opposite* expected sign. It is a *flow change*, so it shares little variance with the slow-moving *level* signals `intangible_value` / `intangible_profitability` (which live in a 0.20-decay `K_int` stock). Vs `rd_productivity` ($\Delta \text{sales} / \text{K_int}$, an *output* measure) it is an *input-commitment* measure that is high precisely *before* sales respond.

Why persistent / not arbitrated. Behavioral fixation on EPS + no capitalised-asset line to correct the anchor + a multi-year, noisy payoff that requires being long names with *deteriorating reported profitability* through the wait.

Prediction. Long high; **moderate**, $|t(\alpha)| \approx 2-3$. A single-industry sort controls the sector innovation cycle; YoY-change noise caps it below "strong." Anchor: Eberhart, Maxwell & Siddique (2004); Lev & Sougiannis (1996); Cohen, Diether & Malloy (2013).

2.2 `rd_conversion` — the profit output: does R&D convert to *margin*? (long high)

Definition. $(\text{gross_income_ltm} - \text{gross_income_ltm}_{(t-12)}) / \text{sales_ltm}_{(t-12)}$ — the YoY change in **gross profit**, scaled by prior-year sales (a stable, always-positive base — deliberately **not `K_int`**).

Predicting logic. `rd_productivity` asks whether R&D produces *sales* — but in software, sales can be bought with discounting or unprofitable land-grab, so revenue is a weak proxy for value created. The economically decisive question is whether R&D converts into **gross profit**: durable, high-margin, pricing-power output (gross income nets out COGS / hosting). Rewarding margin-accretive innovation and penalising low-margin revenue-chasing is precisely the profit-vs-sales distinction the existing book misses. Notably, `rd_productivity` **printed a negative alpha ($t \approx -1.1$) in Experiment 2** — consistent with raw sales-per-R&D rewarding the wrong thing; switching the output measure to *profit* is designed to repair that documented failure.

Why robust. Inherits the robustness of the gross-profitability anomaly (Novy-Marx 2013 — gross margin is the least-manipulable profitability line) applied to the innovation-output question, and is built on large, stable accounting lines (lower noise than change/second-difference factors).

Why not correlated with existing factors. Distinct from `rd_productivity` on **both** axes — numerator (gross *profit* vs sales) and denominator (a safe sales base vs the unstable `K_int`); they diverge for exactly the firms that matter (a firm buying unprofitable growth scores high there, low here). Vs `gross_profitability` ($\text{gross_income} / \text{assets}$, Exp1) it is a *change* normalised by lagged sales, isolating the *increment* rather than the static level. Vs `operating_leverage` ($\Delta \text{opinc} / \Delta \text{sales}$) it uses *gross* not operating income and a *prior-sales* (not Δsales) base.

Why persistent / not arbitrated. Sell-side over-rewards top-line/bookings and under-rewards their *profitability*; gross-margin attribution to R&D vintages is not disclosed (you must construct it); shorting margin-poor revenue-growers is painful while their revenue still visibly grows (the Lev–Sougiannis under-incorporation has survived 25+ years of publication).

Prediction. Long high; **moderate-to-strong**, $|t(\alpha)| \approx 2.5-3$. Our highest-conviction *standalone* of the four — two robust mechanisms (R&D output conversion + the gross-margin channel) on low-noise inputs. Anchor: Lev & Sougiannis (1996); Novy-Marx (2013); Peters & Taylor (2017).

2.3 **rd_stability** — the consistency: is the R&D programme disciplined? (long high)

Definition. $-\text{std}_{36m}(\text{R\&D/sales}) / \text{mean}_{36m}(\text{R\&D/sales})$ per stock (negative trailing-36m coefficient of variation; ≥ 24 valid months required). High (near 0) = a steady, committed R&D programme; very negative = erratic spending.

Predicting logic. Steady R&D is the signature of a governance-disciplined, multi-year innovation programme; volatile intensity is the signature of **real-activities earnings management** — managers cut discretionary R&D to hit EPS targets (Graham, Harvey & Rajgopal 2005: a *majority* of CFOs admit they would). Because R&D is expensed, cutting it boosts reported earnings dollar-for-dollar this quarter while mortgaging future cash flow. A firm that holds intensity steady is signalling it is *not* pulling that lever — higher-quality, more persistent earnings, a dimension the market under-prices.

Why robust. Rests on a permanent agency conflict (managerial myopia under earnings pressure) + permanent expensing. It is a unit-free dispersion measure and the 36-month window makes it slow-moving → **low turnover** → **low transaction cost**.

Why not correlated with existing factors. It is the **only second-moment signal in the entire 15-factor book** — every other factor is a level or a first difference. By construction (dispersion $\div$ mean) it carries *no* information about the *level* of R&D, so it is orthogonal to

$\text{intangible_value} / \text{intangible_profitability} / \text{rd_productivity}$, and it separates cleanly from rd_growth (a one-off spike scores *high* on growth but *low* on stability). Vs accruals (Exp1): accruals catch *accrual-based* manipulation, but cutting a real cash expense is *real-activities* manipulation that leaves accruals untouched (it actually *raises* operating cash flow) — a channel accruals structurally cannot see. It is not beta (that is *return* volatility, not *fundamental-flow* volatility).

Why persistent / not arbitrated. Requires a **per-stock 36-month panel moment** — absent from vendor libraries of point-in-time cross-sectional ratios — and a long, catalyst-free payoff horizon unattractive to event-driven capital. The premium is compensation for *not* being exposed to opportunistic R&D cutting, a risk invisible without explicitly computing the moment.

Prediction. Long high (steady); **weak-to-moderate**, $|\text{t}(\alpha)| \approx 1.5\text{--}2.5$. Earnings-quality / stability premia are real but diffuse, and the monthly t-stat is compressed by the slow payoff; we expected its main value to be *diversification* (near-zero correlation to the rest of the book). Anchor: Graham, Harvey & Rajgopal (2005); Roychowdhury (2006); Bushee (1998); the *quality-stability lineage* (Asness–Frazzini–Pedersen QMJ).

2.4 **innovation_mix** — the composition: build vs sell? (long high)

Definition. $\text{rd_ltm} / (\text{rd_ltm} + \text{sga_ltm})$ — R&D as a share of total discretionary (expensed) spend. Range [0, 1].

Predicting logic. R&D and SG&A are both expensed and both depress current earnings, but they are economically opposite: R&D builds a durable, off-balance-sheet intangible asset; much of SG&A — especially S&M — *rents* current revenue that evaporates when you stop paying. GAAP buries both in one "expensed" bucket, so two firms with identical margins can have radically different *quality of spend*. A firm tilting its discretionary dollar toward R&D accumulates more off-balance-sheet capital per dollar of reported expense; the market sees only the aggregate margin hit, not the build-vs-burn composition underneath.

Why robust. A pure cross-sectional ratio of two large, stable reported line items — no estimation, no window, defined every month for every firm reporting both — encoding a permanent economic distinction (capital-building vs revenue-renting spend).

Why not correlated with existing factors. Vs `gtm_efficiency` ($\Delta\text{sales}/\text{SG\&A}$) it measures the *allocation* between building and selling, not the *productivity* of the selling dollar — different numerators (R&D vs Δsales); the signals diverge for any sales-led but sales-efficient grower. Vs `intangible_profitability` (a profitability *level* over an asset+ `K_int` base) it has no profit/asset/price term at all. The only overlap to monitor is a mild positive tilt with `gross_profitability` (R&D-led firms tend to higher margins).

Why persistent / not arbitrated. Line-item aggregation into operating expense hides the mix from headline screens (most quant pipelines never decompose opex); investors reward visible S&M-driven "land-grab" growth and under-reward quiet product investment; the penalty for sell-tilted growth (margin erosion when S&M can no longer be cut) surfaces years later, beyond the arbitrage horizon. Clean R&D/SG&A splits are noisy *across* industries — the within-software setting is exactly where the edge exists.

Prediction. Long high (R&D-tilted); **moderate**, $|t(\alpha)| \approx 2$. A composition/quality tilt rather than a sharp predictor, with likely mild concavity at the extreme (a very high R&D share can flag a distressed pre-revenue firm that slashed S&M) — so the even-quintile sort should read it better than the linear FM slope. *Anchor:* Peters & Taylor (2017); Enache & Srivastava (2018); Eisfeldt–Papanikolaou (*organisation capital*).

2.5 `rd_intensity` — the level (conventional baseline) (*long high*)

Definition. `rd_ltm / sales_ltm` — the textbook R&D-intensity ratio. Retained **only as the conventional anchor**: it is a *level*, not a behavior, and is the most widely studied R&D signal (Chan, Lakonishok & Sougiannis 2001). Its job here is to let us test the central thesis — *does R&D behavior beat the R&D level?* — head-to-head.

Prediction. Long high; **modest**, $|t(\alpha)| \approx 1\text{--}2$, but flagged ex-ante as the *least* reliable of the five, with explicit downside risk that the post-2016 de-rating of unprofitable, high-spend software could push its industry-neutral alpha *negative* in the recent sub-sample.

Summary of ex-ante predictions

Factor	Dimension	Predicted direction	Predicted $ t(\alpha) $	Conviction
<code>rd_conversion</code>	profit output	long high (+)	2.5–3	highest standalone
<code>rd_growth</code>	flow	long high (+)	2–3	moderate
<code>innovation_mix</code>	composition	long high (+)	~2	moderate
<code>rd_stability</code>	consistency	long high (+)	1.5–2.5	diversifier
<code>rd_intensity</code>	level (baseline)	long high (+)	1–2	low (de-rating risk)

3. How they were tested (Experiment 2 pipeline)

Identical machinery to Experiments 1–2, reused **unmodified** via dependency injection (`main_rd.py` registers `rd_factors` as `sys.modules["factors"]` , so Experiment 1's `quintile.py` / `regression.py` / `cost.py` run against these factors with zero code changes):

- **Point-in-time fundamentals.** R&D, sales, SG&A, assets and gross income are attached to each month-end by a backward as-of merge on `observation_date` (a report enters month t only once it was observable) — the project's standard look-ahead fix.
- **Standardisation.** Each factor is winsorised at 1%/99% and z-scored within each month, relative to the industry mean.
- **Approach 1 — even-quintile sorts.** Five equal-count buckets on the z-score each month; the **next-month** mean return of each, the Q5–Q1 spread, its t-stat and Sharpe.
- **Approach 2 — Fama–MacBeth.** Monthly cross-sectional OLS of next-month return on the z-score; the time-series mean slope and its FM t-stat (full sample and 2016+).
- **Industry-neutral alpha.** The signed dollar-neutral long/short book is regressed on the equal-weighted industry return; **α (and $t(\alpha)$) is the headline** — return not explained by industry exposure. Average monthly **turnover cost** (one-way, charged on traded weight only) is reported alongside.
- **Redundancy.** Experiment 3's panel-agnostic `factor_correlation.run` measures each R&D factor's R^2 against (a) the 9 Exp1 general factors and (b) the 6 Exp2 software factors; `rd_diagnostics.py` adds the 5×5 cross-correlation among the R&D factors themselves.

Outputs: `RD/{factor_panel.csv, quintile/..., regression/..., factor_correlation/...}`.

4. Results & comparison to the hypotheses

4.1 Headline — industry-neutral long/short alpha (canonical long-high direction)

Long-short quintile strategy regressed on the industry return ($ls_t = \alpha + \beta \cdot \text{industry}_t + \varepsilon$; α = industry-neutral monthly return, t-stat tests $\alpha \neq 0$)												
Factor	Family	L/S direction	Alpha (monthly)	Alpha t-stat	Sharpe (ann.)	β -neutral Sharpe	Avg cost (monthly)	n	Alpha (2016+)	Alpha t (2016+)	Sharpe (2016+)	β -neutral Sh (2016+)
rd_growth	R&D growth (flow change)	Q5-Q1	-0.3765%	-1.39	-0.05	-0.27	-0.0677%	324	-0.2367%	-0.66	+0.00	-0.21
rd_conversion	R&D profit conversion (output)	Q5-Q1	+0.1504%	+0.66	+0.24	+0.13	-0.0749%	324	+0.2450%	+0.66	+0.48	+0.21
rd_stability	R&D stability (commitment consis	Q5-Q1	+0.8248%	+3.86	+0.37	+0.76	-0.0338%	313	+1.1065%	+3.54	+0.62	+1.15
innovation_mix	Innovation mix (build vs sell)	Q5-Q1	+0.0278%	+0.11	+0.27	+0.02	-0.0250%	336	-0.1307%	-0.40	-0.18	-0.13
rd_intensity	R&D intensity (level; baseline)	Q5-Q1	-0.4336%	-1.46	+0.18	-0.28	-0.0262%	336	-0.7976%	-1.99	-0.21	-0.65

Shaded alpha t-stats: $|t| \geq 1.65$ (10%), darker $|t| \geq 2.0$ (5%). Sharpe = annualised Sharpe of the L/S book. Avg cost = mean monthly turnover cost (one-way, traded weight only). 2016+ columns re-estimate α on the past decade only.

Cross-sectional factor premia: mean monthly β and Fama-MacBeth t-stat
(next-period return on factor z-score; β = return per 1 std of factor)

Factor	Family	Mean β (full)	FM t (full)	Mean β (2016+)	FM t (2016+)
rd_growth	R&D growth (flow change)	-0.0009	-0.88	+0.0001	+0.11
rd_conversion	R&D profit conversion (output)	+0.0008	+1.12	+0.0000	+0.04
rd_stability	R&D stability (commitment consist)	+0.0021	+2.24	+0.0028	+1.89
innovation_mix	Innovation mix (build vs sell)	+0.0010	+1.22	-0.0005	-0.50
rd_intensity	R&D intensity (level; baseline)	-0.0009	-0.94	-0.0017	-1.23

Shaded FM t-stats: $|t| \geq 1.65$ (10%), darker $|t| \geq 2.0$ (5%)

Factor	Gross Q5–Q1 /mo			FM t (full)	α 2016+	$t(\alpha)$ 2016+	FM t 2016+	Sharpe	Avg cost (pp/mo)
rd_stability	+0.49%	+0.82%	+3.86	+2.24	+1.11%	+3.54	+1.89	0.37	0.034
rd_conversion	+0.29%	+0.15%	+0.66	+1.12	+0.24%	+0.66	+0.04	0.24	0.075
innovation_mix	+0.38%	+0.03%	+0.11	+1.22	-0.13%	-0.40	-0.50	0.27	0.025
rd_growth	-0.07%	-0.38%	-1.39	-0.88	-0.24%	-0.66	+0.11	-0.05	0.068
rd_intensity (baseline)	+0.37%	-0.43%	-1.46	-0.94	-0.80%	-1.99	-1.23	0.18	0.026

(α and $t(\alpha)$: industry-neutral monthly alpha of the Q5–Q1 book; FM t: Fama–MacBeth t-stat of the cross-sectional premium. Bold = $|t| \geq \sim 2$.)

4.2 Scorecard — predicted vs realised

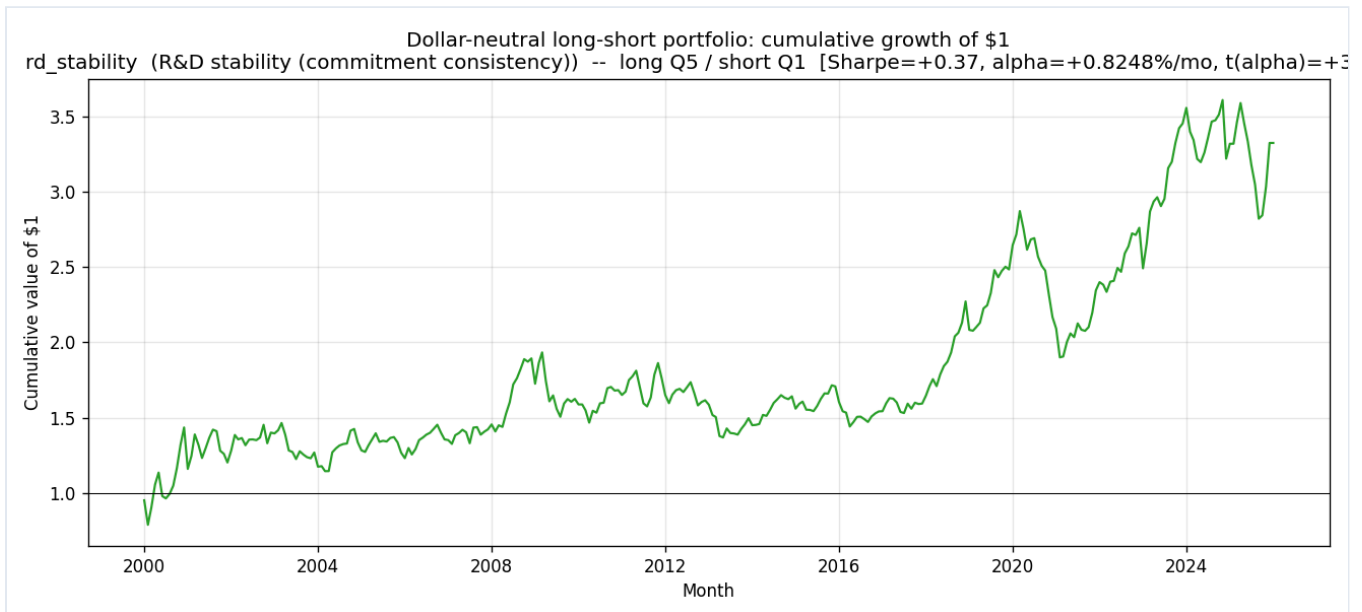
Factor	Predicted	Realised (full-sample $t(\alpha)$)	Verdict
rd_stability	long high, $ t \approx 1.5$ –2.5	+3.86 (2016+: +3.54)	✅ Confirmed & exceeded — direction right, magnitude <i>well above</i> prediction; the standout.
rd_conversion	long high, $ t \approx 2.5$ –3	+0.66 (FM +1.12)	🟡 Partial — sign correct, magnitude far below prediction; only directionally supportive.
innovation_mix	long high, $ t \approx 2$	+0.11	🟡 Partial — sign correct but ~zero industry-neutral alpha; the raw spread is mostly beta.
rd_growth	long high, $ t \approx 2$ –3	-1.39	❌ Failed — wrong sign (mildly negative), insignificant.
rd_intensity (baseline)	long high, $ t \approx 1$ –2 (de-rating risk)	-1.46 (2016+: -1.99)	❌ Reversed — <i>as flagged</i> : significantly negative post-2016.

4.3 The one big highlight: **rd_stability** works, and it is the right *kind* of factor

rd_stability is the **only** factor here with a significant industry-neutral alpha, and it is significant on every cut: full-sample $t(\alpha) = +3.86$ ($\alpha = +0.82\%/mo$), Fama–MacBeth $t = +2.24$, and — unusually for this project — it **strengthens after 2016** ($\alpha = +1.11\%/mo$, $t(\alpha) = +3.54$), where even strong Experiment-2 factors like

intangible_value fade. Three features make it a genuinely high-quality signal, not a fluke:

1. **Its alpha exceeds its gross spread** (+0.82% α vs +0.49% raw Q5–Q1). That can only happen if the book carries *negative* industry beta — i.e. the stable-R&D leg is **defensive**, so hedging the industry *adds* return. By contrast **rd_intensity**, **innovation_mix** and **rd_conversion** all have gross spread > α , meaning their high-R&D long legs load *positively* on the industry and their apparent raw premium is largely beta, not alpha.
2. **It is cheap to run** — 0.034 pp/month average turnover cost (the 36-month window makes it slow-moving), so the net alpha is $\approx 0.79\%/mo$. Economically tradable.
3. **It is distinct** (see §4.5): only $\sim 14\%$ of it is spanned by the general factors and $\sim 19\%$ by the software factors — the rest is new information.

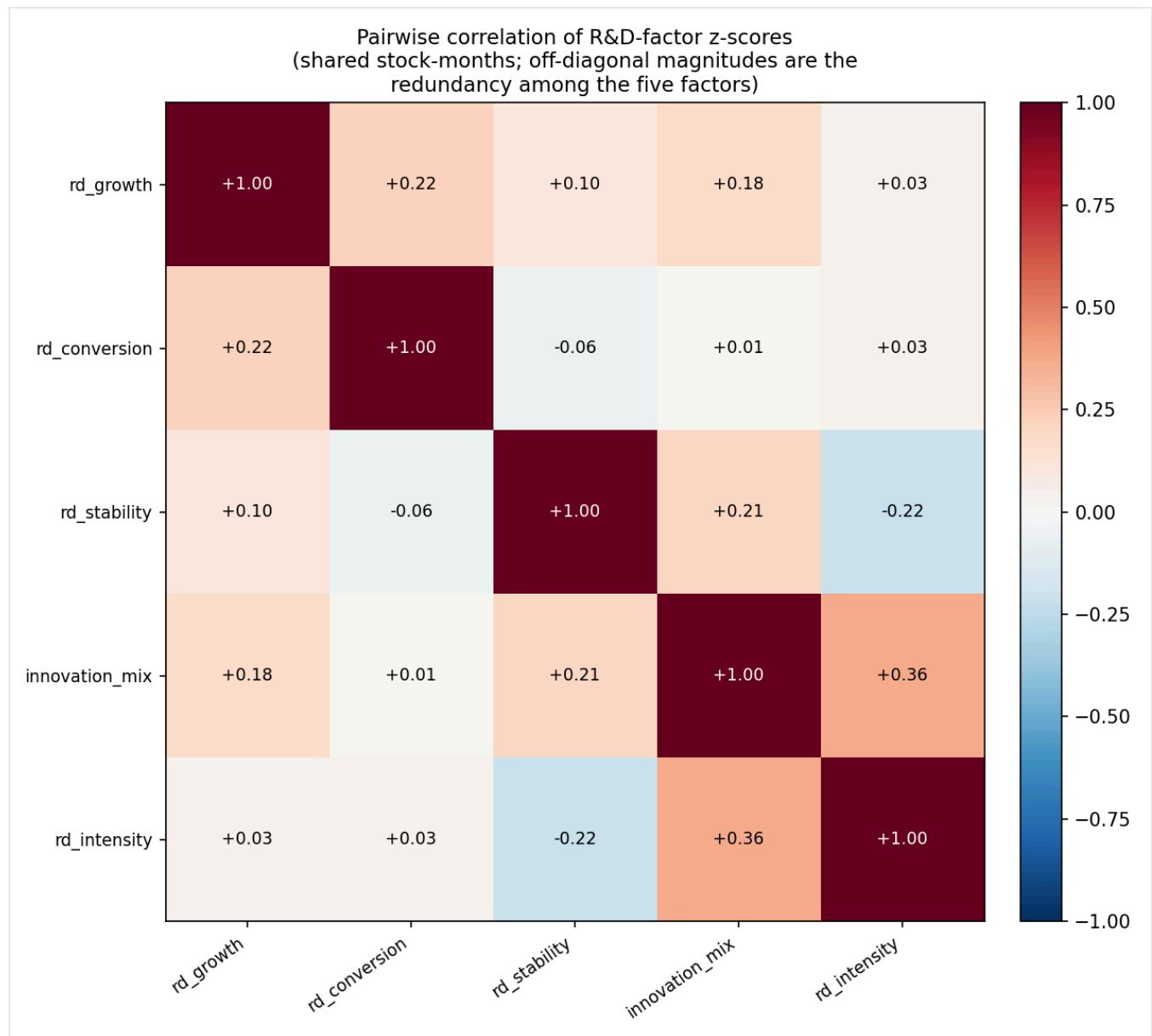


We **under-predicted** it: pitched ex-ante as a weak-to-moderate *diversifier* ($|t| \approx 1.5\text{--}2.5$), it came in as the strongest factor in the set. The interpretation: within an industry where *everyone* spends heavily on R&D, *how reliably* a firm sustains that spend separates disciplined compounders from firms managing earnings via the R&D lever far more sharply than we expected — and the market pays for that discipline, increasingly so post-2016.

4.4 The thesis is vindicated by the baseline's failure

The central thesis was *behavior beats level*. The data delivers the cleanest possible confirmation: the **behavioral** consistency factor earns $t(\alpha) = +3.86$, while the **conventional level** (**rd_intensity**) earns $t(\alpha) = -1.46$ full-sample and a *significantly negative* **–1.99 post-2016**. Within software, high R&D **spending** is, if anything, a *negative* on an industry-neutral basis recently — exactly the post-2016 de-rating of unprofitable, high-burn software we flagged as the baseline's downside risk. Spending a lot on R&D is not rewarded; spending it *consistently* is. That is the whole point of looking at behavior rather than level.

4.5 Distinctness — the "not correlated with existing factors" claim, measured



Redundancy of the R&D factors with the existing book
JOINT R² = fraction of each R&D factor spanned by ALL of that set's factors together (lower = more distinct)

R&D factor	JOINT R ² vs Exp1 general	top general explainer	JOINT R ² vs Exp2 software	top software explainer
rd_growth	0.083	asset_growth (R2=0.042)	0.052	gtm_efficiency (R2=0.019)
rd_conversion	0.105	asset_growth (R2=0.064)	0.088	gtm_efficiency (R2=0.042)
rd_stability	0.141	gross_profitability (R2=0.080)	0.195	intangible_profitability (R2=0.128)
innovation_mix	0.010	gross_profitability (R2=0.002)	0.089	rd_productivity (R2=0.044)
rd_intensity	0.072	accruals (R2=0.039)	0.112	intangible_profitability (R2=0.093)

Benchmark: the existing rd_productivity sits at JOINT R²=0.03 vs the general set. Shading: R² ≥ 0.03, 0.10, 0.25.

- Distinct from each other.** The largest pairwise z-score correlation among the five is **0.36** (`innovation_mix` – `rd_intensity`); `rd_stability` vs the rest sits at $|p| \leq 0.22$ and `rd_conversion` is near-orthogonal to everything ($|p| \leq 0.22$). The five span genuinely different dimensions.

- **Distinct from the existing book.** JOINT R^2 (the share of each R&D factor explained by *all* of a set's factors together) is low across the board — `innovation_mix` 0.010 vs the general set, `rd_growth` 0.083 / 0.052, `rd_conversion` 0.105 / 0.088, `rd_intensity` 0.072 / 0.112. Even the *most*-correlated, `rd_stability`, leaves **86%** unexplained by the general set and **81%** by the software set; its largest single explainer is `gross_profitability` / `intangible_profitability` — i.e. it is *partly* a quality signal, but four-fifths of it is new. The empirical "not redundant" claim holds: every factor clears the existing `rd_productivity` benchmark (JOINT $R^2 \approx 0.03$) only modestly and none is spanned.

4.6 Why the two "partials" landed where they did

- `rd_conversion` — sign correct and the best-behaved of the non-stability behavior factors (positive raw spread +0.29%/mo, FM $t = +1.12$, post-2016 Sharpe 0.48), but the industry-neutral alpha is only +0.66 t . It *did* avoid `rd_productivity`'s outright negative result — switching the output measure from sales to gross profit removed the perverse reward to low-margin growth, as designed — but it did not reach the 2.5–3 t we predicted. The profit-conversion premium is real-but-diffuse at monthly frequency in this single industry; lumpy YoY gross-profit changes add noise.
- `innovation_mix` — sign correct, raw spread +0.38%/mo (FM $t = +1.22$), but ~zero industry-neutral alpha: the build-vs-sell tilt is real in *gross* terms but is almost entirely an industry-beta exposure (R&D-tilted names are higher-beta), so hedging the industry leaves nothing. Consistent with our ex-ante note that its predictive content is a soft quality tilt rather than a sharp alpha source.

4.7 Why `rd_growth` failed

The Eberhart-type "R&D increase" premium is a *cross-industry* result; **within** a single R&D-intensive industry it does not survive at monthly frequency — here it is mildly *negative*. The most plausible reading: in software, ramping R&D/assets coincides with cash-burning over-investment and acquisitive asset growth (the very thing the investment anomaly penalises), and the single-industry sort strips out the sector-level innovation signal that drives the original effect. The lesson reinforces §4.4: the *level/flow* of R&D spend is not the edge — its *discipline* is.

4.8 Robustness / look-ahead audit

Because `rd_stability` is the headline result — and because this project has a documented history of look-ahead bugs — it was put through an **independent adversarial audit** whose explicit mandate was to *refute* it. The verdict was **SURVIVES**.

- **Look-ahead: clean.** Fundamentals are attached on `observation_date`, not the fiscal `date_fundamental` (which precedes publication 42.5% of the time and *would* leak). Across all 152,612 attached stock-months, **zero** used an observation date later than the month-end it was attached to. The 36-month rolling window is strictly backward-looking; `next_return` is verified to be the realised month $t+1$ return (the factor never sees the return it predicts); the z-score is computed within month t only.
- **Not a mechanical artefact.** The stability score correlates **+0.125** with the *number* of distinct R&D updates in the window — i.e. firms that report *more* genuine updates look *more* stable, the **opposite** of a "less-frequent-reporting-looks-smoother" artefact. There is no microcap tilt (the stable long leg holds the *largest* names, median \$2.87B, vs \$0.52B in the short leg), the spread survives a **size double-sort** (Q5–Q1 = +1.02% / +0.60% / +0.53% per month across size terciles, positive everywhere), and dropout rates are flat across quintiles (no differential survivorship). The fact that α (**+0.87%**) **exceeds**

the raw spread (+0.49%) is legitimate: the book has a genuine *negative* industry beta (−0.33), so hedging the industry cuts residual volatility (3.81% vs 4.52%) and lifts the intercept — exactly how a market-model alpha should behave.

- **Independently reproduced.** A standalone re-implementation (raw feathers, no project code, point-in-time on `observation_date ≤ month-end`) reproduced the numbers almost exactly: full-sample α **+0.87%/mo, $t = 3.98$** (project: +0.83%, $t = 3.86$); 2016+ α **+1.11%/mo, $t = 3.50$** (project: +1.11%, $t = 3.54$). Positive in every sub-period (1999–07, 2008–15, 2016–26).
- **One honest caveat.** The universe is selected on the *current* GICS classification (the security master carries no time-varying industry) — a universe-membership look-ahead **shared by the entire project**, not specific to this factor, and one that cannot manufacture the *within-universe* cross-sectional stability ranking's predictive power.

4.9 Caveats & limitations

- **One industry, one regime of accounting.** Results are within-Software & Services; the expensing mechanism generalises, but the magnitudes do not necessarily.
- **rd_stability is partly a quality/low-vol signal.** ~13–19% of it is shared with profitability factors; its alpha is industry-neutral but we have not orthogonalised it against a full quality composite. Part of its premium may be the documented quality-stability premium expressed through the R&D lens (which is itself the point — it is a *novel construction* of that premium — but it is not 100% orthogonal).
- **Monthly forward-filled LTM ratios.** Fundamentals update ~quarterly and are carried forward; the rolling moment in `rd_stability` is computed on that step-function series. The independent audit (§4.8) checks this is a feature, not an artefact.
- **Coverage.** ~75% of software stock-months report R&D; firms with no R&D are (correctly) absent from these factors, so the cross-section is the R&D-reporting subset.

5. Conclusion & implications for Experiment 3

The exercise produced **one clear keeper**, one near-miss, and a clean validation of the guiding thesis:

- **Promote `rd_stability` into the multifactor model.** It is significant ($t(\alpha) = 3.86$), cheap (0.034 pp/mo), distinct ($\text{JOINT } R^2 \leq 0.20$), defensive (negative industry beta), and — rare in this project — **post-2016 robust** ($t(\alpha) = 3.54$). It is the project's first *second-moment* signal and adds a diversification axis the existing book lacks. It belongs alongside `buyback_quality` in Experiment 3's $|\alpha| > 5$ / low-correlation screen-adjacent set (it clears the alpha-significance and low-correlation bars even if not the strict $|t| > 5$ cut-off used there).
 - **Hold `rd_conversion` as a watch-list candidate.** Directionally correct and it repaired `rd_productivity`'s sign; worth re-testing with a longer holding period or combined with `rd_stability`.
 - **Drop `rd_growth` and the `rd_intensity` baseline as standalone signals.** The level/flow of R&D spend is not rewarded within software; `rd_intensity` is *significantly negative* post-2016.
 - **Headline lesson:** in an industry defined by R&D, *how* a company spends — consistently, and converting to margin — carries cross-sectional alpha that *how much* it spends does not. Behavior beats level.
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